POWER

File: power.kicad_sch

NRF

File: nrf.kicad_sch

MSP

File: msp.kicad_sch

TX

File: tx.kicad_sch

RX

File: rx.kicad_sch

CMUT

File: cmut.kicad_sch

PPG

File: ppg.kicad_sch

Sheet: /
File: BAgmenzi.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 1/8

U1

MSP430FR5043IRGCT

AVCC1	1	AVCC1			
DVCC1	18	DVCC1			
DVCC3	49	DVCC3			
PVCC	56	PVCC			
PVCC	57	PVCC			
VGA_GAIN_RC_EN	3	P1.0_UCA1CLK_TA1.0_A0_C0_VREF-_VeREF-			
VGA_GAIN_CS	4	P1.1_UCA1STE_TA4.0_A1_C1_VREF+_VeREF+			
	19	P1.2_UCA1SIMO_UCA1TXD_TA1.0_A8_C8			
VRX_EN	20	P1.3_UCA1SOMI_UCA1RXD_TA1.1_A9_C9			
	25	P1.4_TB0.4_UCB0STE_A2_C2			
	26	P1.5_TB0.5_UCB0CLK_A3_C3			
	23	P1.6_UCA3STE_UCB0SIMO_UCB0SDA			
UUPS_TRIG_IN	24	P1.7_USSTRG_UCA3CLK_UCB0SOMI_UCB0SCL			
BSL.UART_TX	21	P2.0_UCA1CLK_UCA3SIMO_UCA3TXD			
BSL.UART_RX	22	P2.1_UCA1STE_UCA3SOMI_UCA3RXD			
VGA_GAIN_RC_SINK	2	P2.2_COUT_UCA0CLK_A14_C14			
LP_FILT_EN	27	P3.1_TA1CLK_TB0.1_MTIF_OUT_IN			
HV_POS_EN	50	P3.3_MCLK_TB0.3_XPB1			
HV_NEG_EN	51	P3.4_SMCLK_DMAE0			
HV_PWR_EN	52	P3.5_ACLK_COUT			
	28	P4.0_RTCCLK_TA4.1_MTIF_PIN_EN			
LED_MSP430	29	P4.1_UCA0CLK_TB0.4_UCA3SOMI_UCA3RXD			
	30	P4.2_UCA0STE_TB0.5_UCA3SIMO_UCA3TXD			
	31	P4.3_UCA0SIMO_UCA0TXD			
	32	P4.4_UCA0SOMI_UCA0RXD			
NRF_SPI.MOSI	34	P5.0_TB0.0_UCA2SIMO_UCA2TXD			
NRF_SPI.MISO	35	P5.1_TB0.1_UCA2SOMI_UCA2RXD			
NRF_SPI.SCLK	36	P5.2_TB0.2_UAC2CLK			
NRF_SPI_CS	37	P5.3_TB0.3_UCA2STE			
VGA_SPI.CLK	38	P5.4_TA0.0_UCB1CLK_TA4.0			
VGA_SPI.MOSI	39	P5.5_TA4.1_UCB1SIMO_UCB1SDA			
VGA_SPI.MISO	40	P5.6_TB0OUTH_UCB1SOMI_UCB1SCL			
LINK_READY	41	P5.7_TA0.2_UCB1STE			
DATA_READY	42	P6.0_TA0CLK_COUT			
PULS_HIZ_IN	47	P6.2_TB0CLK			
SPARE_PIN	43	P6.4_MCLK			
	44	P6.5_SMCLK			
TRIG_OUT	45	P6.6_ACLK			
	46	P7.0_TA1.0_TA1.2_XPB0			
			PJ.0_UAC2CLK_SRSCG1_DMAE0_C10_TDO	13	
			PJ.1_UCA2STE_SRSCG0_TA4CLK_C11_TDI_TCLK	14	
			PJ.2_UCA2SIMO_UCA2TXD_SROSCOFF_TB0OUTH_C12_TMS	15	
			PJ.3_UCA2SOMI_UCA2RXD_SRCPUOFF_TB0.6_C13_TCK	16	
			PJ.4_LFXIN	6	LFXIN
			PJ.5_LFXOUT	7	LFXOUT
			PJ.6_HFXIN_USSXT_BOUT	9	HFXIN
			PJ.7_HFXOUT	10	HFXOUT
			CH0_IN	60	CH0_IN
			CH0_OUT	59	CH0_OUT
			CH1_IN	53	
			CH1_OUT	54	
			USSXTIN	62	USSXTIN
			TEST_SBWCLK	11	JTAG.TEST
			RST__NMI_SBWTDIO_N	12	JTAG.RESET_N
			USSXTOUT	63	USSXTOUT
			EP	65	GND
			AVSS1	64	
			AVSS2	5	
			AVSS3	8	GND
			AVSS4	61	
			DVSS1	17	
			DVSS2	33	
			DVSS3	48	
			PVSS	55	GND
			PVSS	58	

C1 1n: CH0_IN — SIG_FILT_IN (2,3)

CH0_OUT — PULSE_OUT (2,4)

R2 1M: CH1_IN — GND

R3 1M: CH1_OUT — GND

JTAG.RESET_N, JTAG.TEST — JTAG{RESET_N, TEST} (2)

2,3 VGA_GAIN_CS — VGA_GAIN_CS

2,3 VGA_GAIN_RC_EN — VGA_GAIN_RC_EN

2,3,7 VRX_EN — VRX_EN

2 BSL{UART_RX, UART_TX} — BSL.UART_TX, BSL.UART_RX

2,3 VGA_GAIN_RC_SINK — VGA_GAIN_RC_SINK

2,3 LP_FILT_EN — LP_FILT_EN

2,4 HV_POS_EN — HV_POS_EN

2,4 HV_NEG_EN — HV_NEG_EN

2,4 HV_PWR_EN — HV_PWR_EN

2,6 NRF_SPI{SCLK, MOSI, MISO} — NRF_SPI.MOSI, NRF_SPI.MISO, NRF_SPI.SCLK

2,6 NRF_MSP_CS — NRF_SPI_CS

2,3 VGA_SPI{SCLK, MOSI, MISO} — VGA_SPI.MISO, VGA_SPI.CLK, VGA_SPI.MOSI

2 NRF_LINK_READY — LINK_READY

2 NRF_DATA_READY — DATA_READY

2,5 SPARE_PIN — SPARE_PIN

2,4 PULSER_HIZ_IN — PULS_HIZ_IN

2,6 UUPS_TRIG_IN — UUPS_TRIG_IN

TRIG_OUT — TP1 TestPoint

add possibility to program through UART bootloader: described in TI's FRAM bootloader guide, SLAU550

JTAG-SPY-wire

J1 SMP: 7 BSL.UART_RX; 6 BSL.UART_TX; 5; 4 GND; 3 JTAG.RESET_N; 2 JTAG.TEST; 1 DVDD_3.3

DVDD_3.3 — R1 47k — JTAG.RESET_N — C2 1n — GND

use pinheaders on BREAKAWAY-> directly connect jumper cables from pin headers to MSP FET.

Stecker muss noch ausgewählt werden.

HF and LF Oscillators

C13 12p — R4 22 — HFXOUT

X1 FY0800018 8MHz: GND, GND, 1, 3

C14 12p — HFXIN

C16 12p — LFXOUT

X2 ABS07-32.768KHZ-9-T 32.768kHz: C18 12p — LFXIN

GND

US-subsystem oscillators

GND — C15 12p — R5 22 — USSXTOUT

Y1 AWSCR-8.00CV-T 8MHz: 1, 2, 3

C17 12p — USSXTIN

Debug LED: keep or drop...

LED_MSP430 — D1 2.65V Green — R6 22 — GND

bypass caps and power supply filters

DVDD_3.3 — L1 600Ohm — C3 47u, C4 47u, C5 1n, C6 1n — GND, GND, PVCC

2,5,6,7 DVDD_3.3 — DVDD_3.3

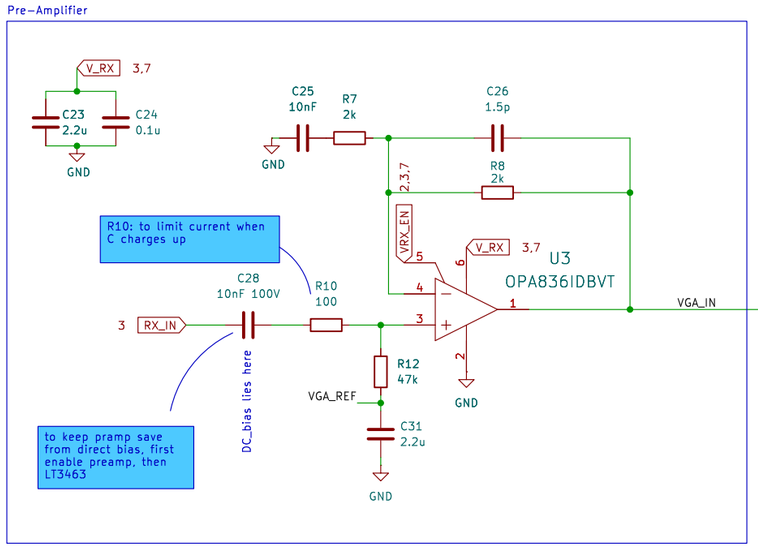
DVDD_3.3 — L2 600Ohm — C7 2.2u, C10 0.1u — AVCC1 — GND

DVDD_3.3 — L3 600Ohm — C8 2.2u, C11 0.1u — DVCC1 — GND

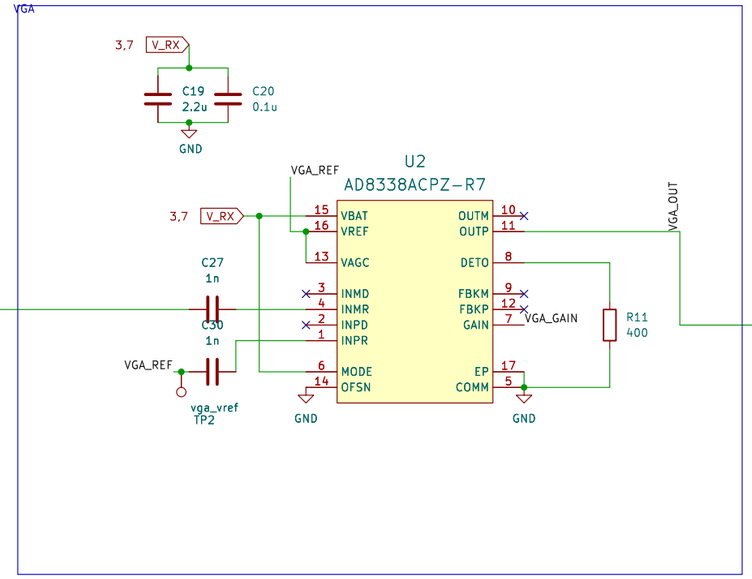
DVDD_3.3 — L4 600Ohm — C9 2.2u, C12 0.1u — DVCC3 — GND

Sheet: /MSP/		
File: msp.kicad_sch		
Title:		
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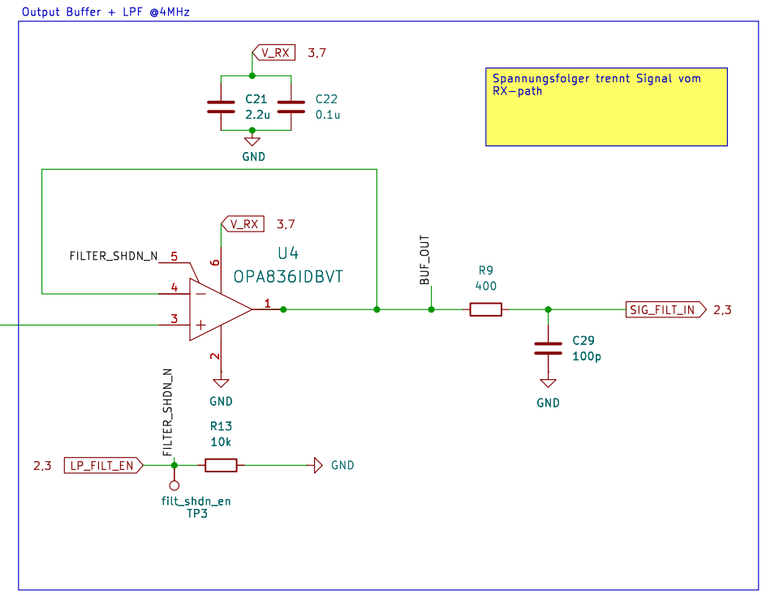
Pre-Amplifier



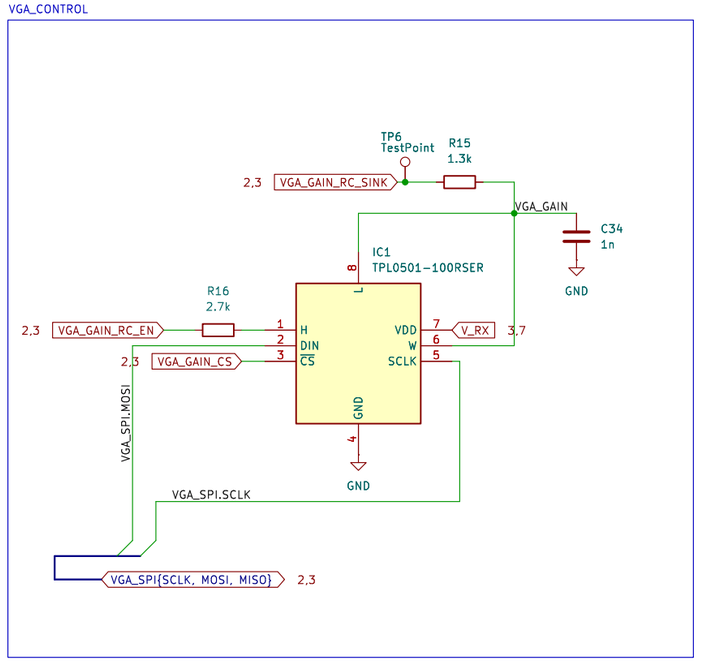
VGA



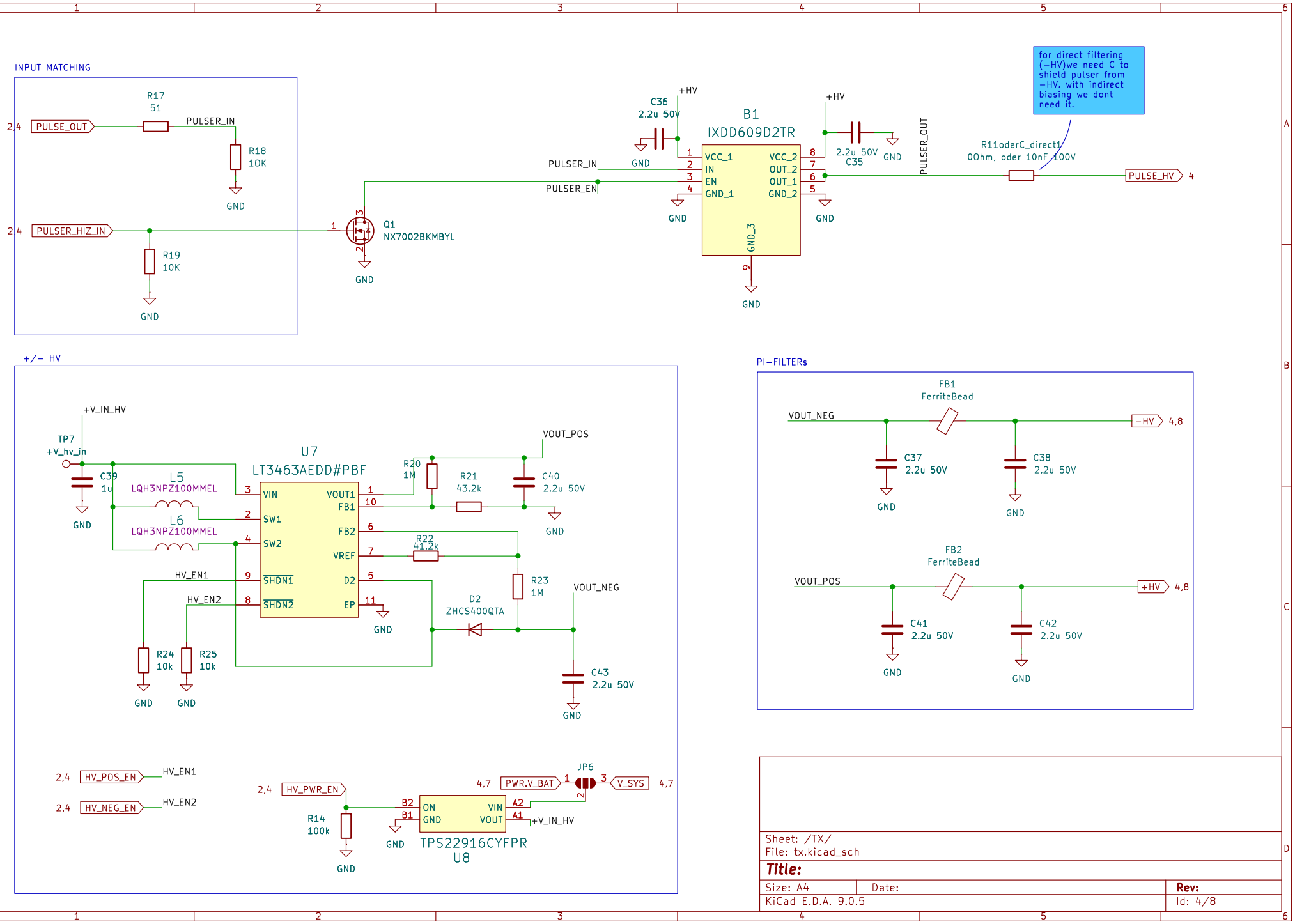
Output Buffer + LPF @4MHz



VGA_CONTROL



Sheet: /RX/		
File: rx.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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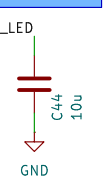
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File: tx.kicad_sch

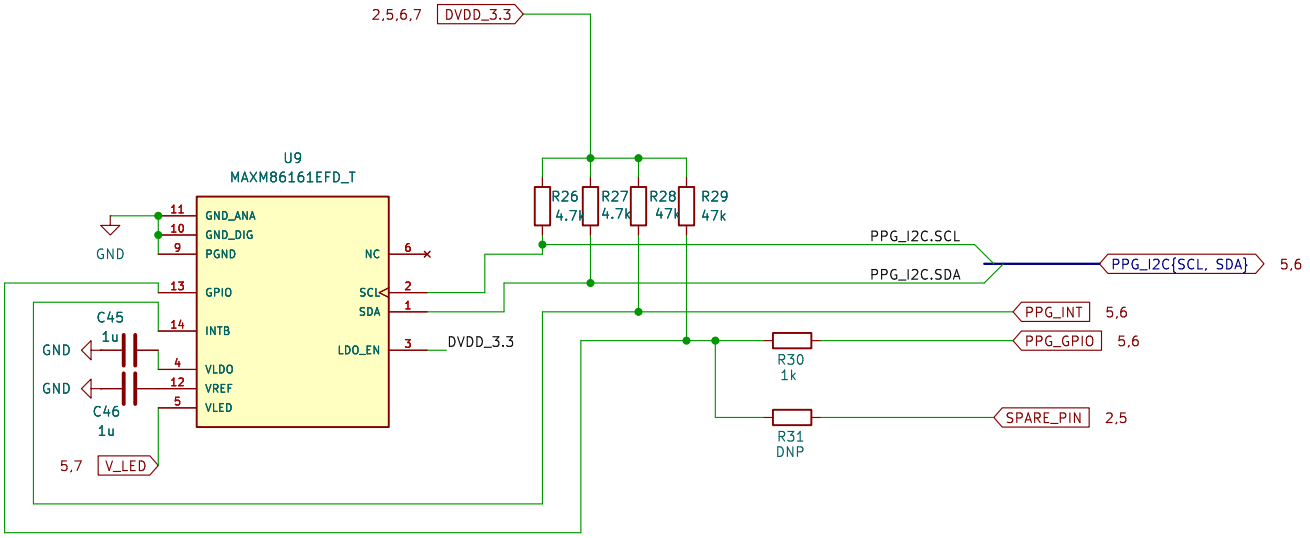
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 4/8

layout: PPG direkt hinter CMUT.
Alternativen mitdenken

VLED based on system application and LED configs used.
V_LED=4.4V

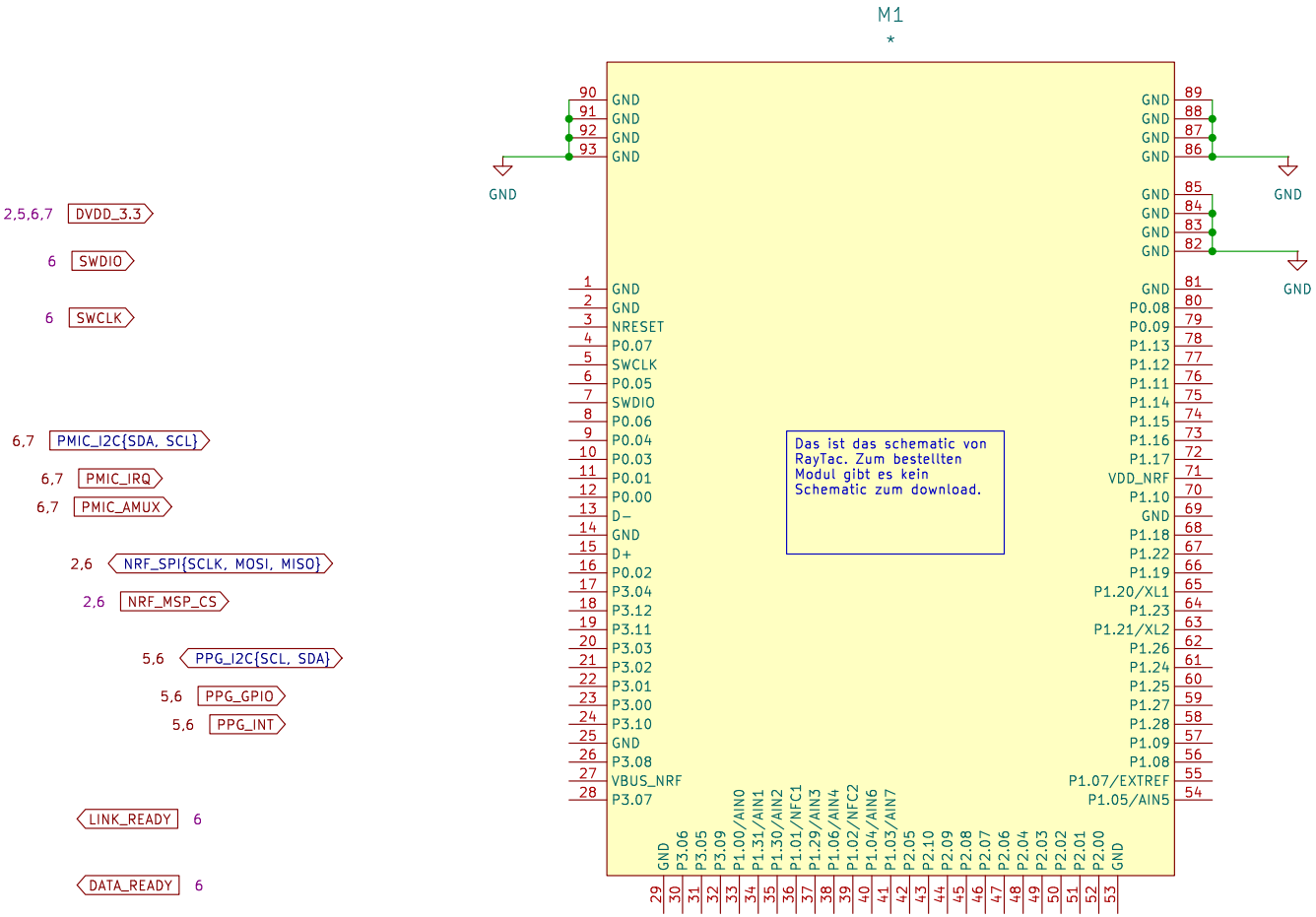




nRF as central clock. it triggers the MSP and the MAXM at the same time, the timedelay until it measures is constant and can then be calibrated.

Alternatively connect it to MSP spare pin and trigger MAXM when pulse is triggered. time delay will be within microseconds

Sheet: /PPG/		
File: ppg.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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2,5,6,7 DVDD_3.3

6 SWDIO

6 SWCLK

6,7 PMIC_I2C{SDA, SCL}

6,7 PMIC_IRQ

6,7 PMIC_AMUX

2,6 NRF_SPI{SCLK, MOSI, MISO}

2,6 NRF_MSP_CS

5,6 PPG_I2C{SCL, SDA}

5,6 PPG_GPIO

5,6 PPG_INT

LINK_READY 6

DATA_READY 6

6 MSP_TRIG

2,6 UUPS_TRIG_IN

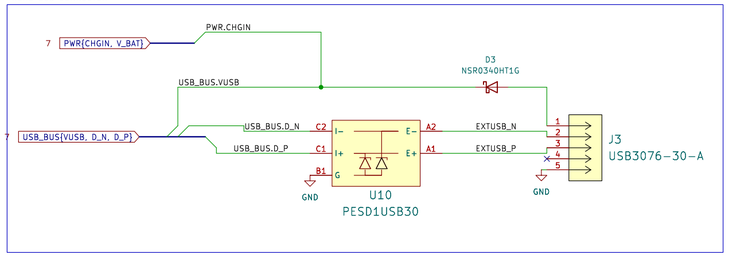
PROGRAMMING CONNECTORS

6,7 soft_RST

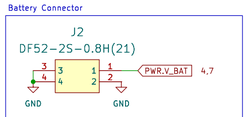
soft reset vom PMIC

Sheet: /NRF/		
File: nrf.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 6/8

MICRO USB CONNECTOR

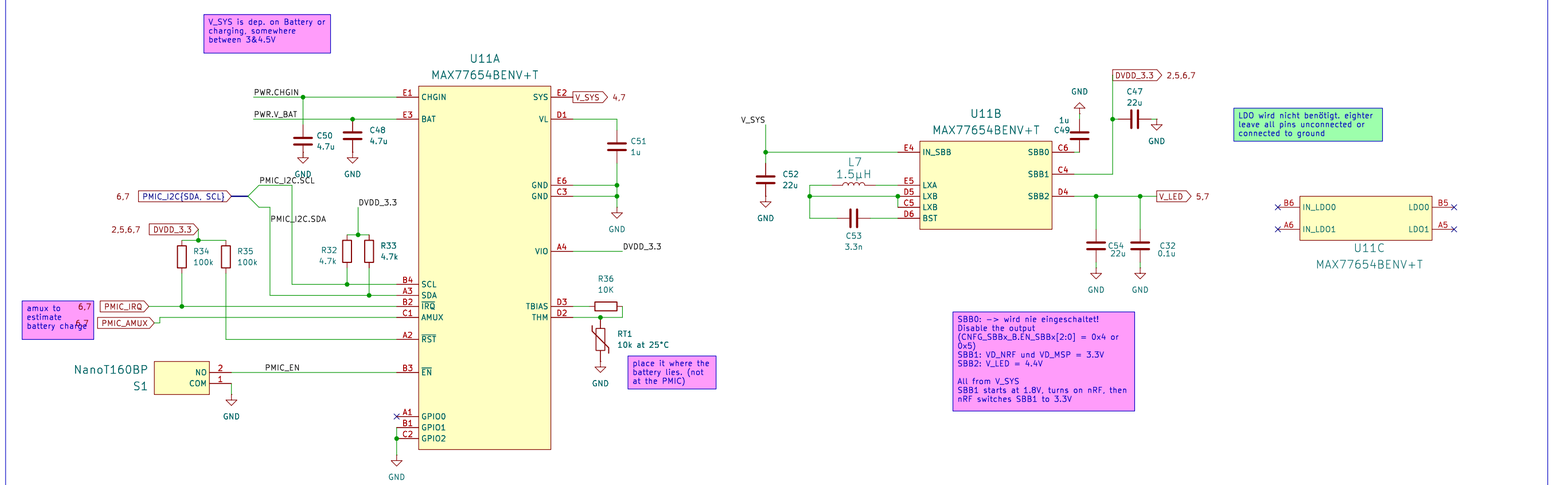


Battery Connector

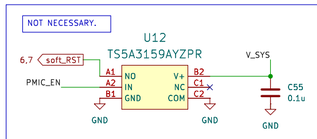


HV POWER IS ON TX PAGE

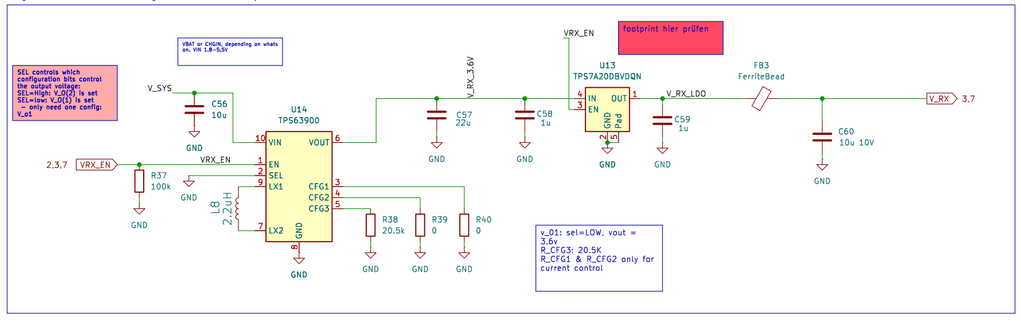
PMIC

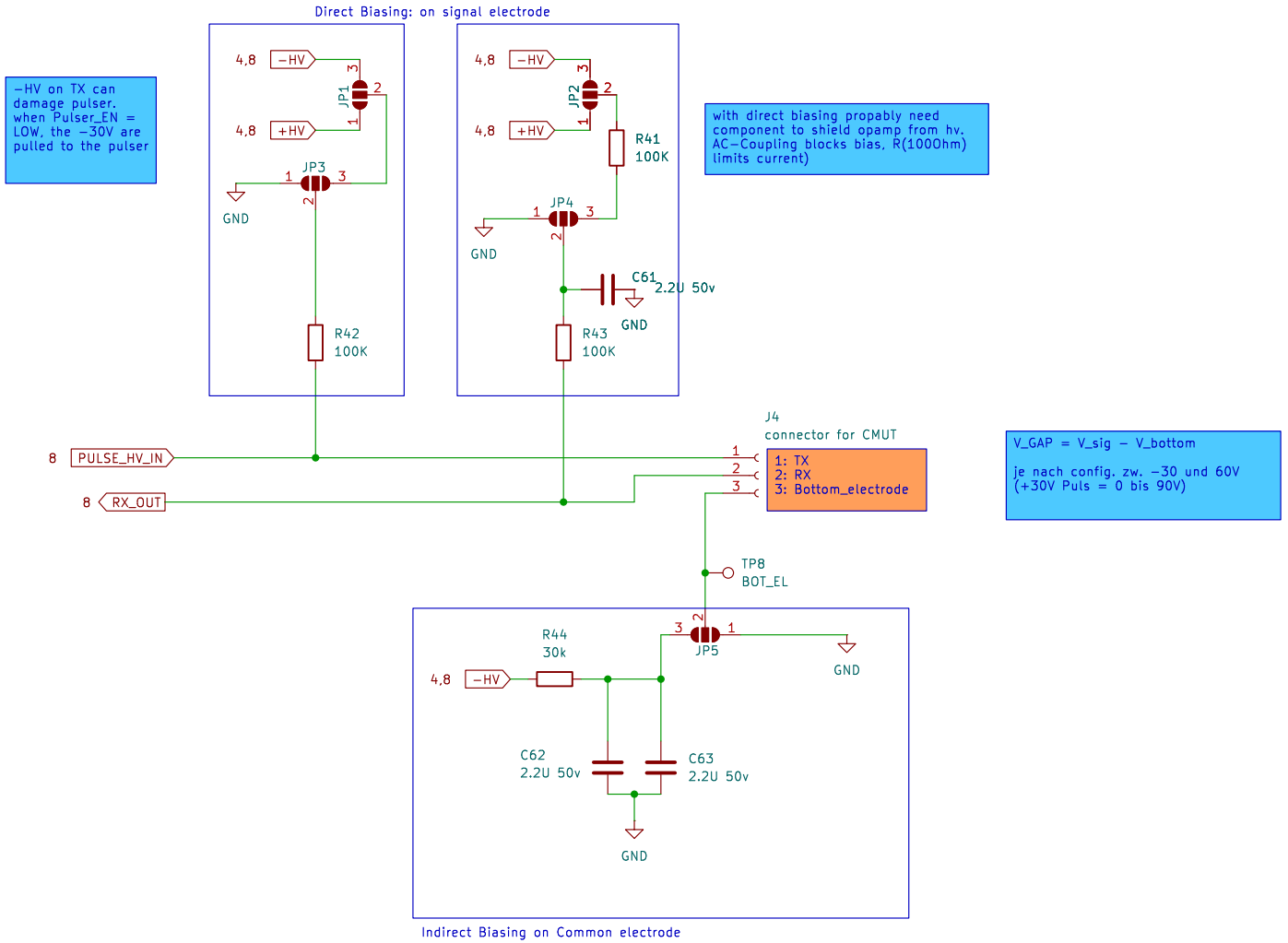


NOT NECESSARY.



to generate V_RX rail: TPS63900 generates 3.6V from V_sys and TPS7A20 LDO cleans it to 3.3V





Sheet: /CMUT/	
File: cmut.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 9.0.5	Id: 8/8